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The lattice width and quasi-straightness in digital spaces

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Abstract

Digital Straightness is a fundamental geometric characteristic in computer science. There have been a lot of equivalent definitions in the past. In the present study, we base our work on the notion of lattice width which is related to the notion of direction of a digital set. Our approach not only applies to thin digital curves but also to thick digital sets. We identify a criterion to detect digital curves that can be considered as nearly straight and we call them quasi-straight digital segments. Experiments are provided on raw data sets with no preprocessing. The results are promising.

1. Introduction

We refer to the book [12] for the notion of Digital Straightness. Previous studies on the mathematical structures of Digital Straightness have served as a basis for many works in digital geometry but are not so popular in pattern recognition. The main reason for this, is that digital straight lines and segments are very rigid structures. It is a mathematical advantage but on irregular shapes, only short segments might be constructed in practice. Previous works [3, 6] extended the notion of DSS so to be able to tolerate some noise in the data either by relaxing the arithmetical constraints or by moving to a geometric characterization. However, the former only applies to thin objects and the latter lost all connections with arithmetics [7]. Hence, in this paper, we propose to fill the gap in order to both retain the arithmetical foundation of DSS and the capability to work with thick digital objects.

The paper is organized as follows. In a first section, we give all necessary definitions. This is followed by a description of a previously published algorithm that has the advantage of being extensible to the incremental case and of having a precise geometrical interpretation. We then provide a definition of quasi-straightness and give some experiments in next section. We end the paper by some conclusions.

2. Definitions

We first review some definitions from algorithmic number theory. Definitions are taken from [2].

Let K be a set of points of $\mathbb{R}^d$ and Λ a lattice included in $\mathbb{R}^d$. Without loss of generality, we suppose both structures to be full-dimensional. The *width* of K along a direction $c \neq 0$ in $\mathbb{R}^d$ is defined as

$$\omega_c(K) = \max\left\{c^T x \mid x \in K\right\} - \min\left\{c^T x \mid x \in K\right\} \quad (1)$$

The *lattice width* of K is the minimal value of the width of K among the directions belonging to the dual lattice Λ^*,

$$\omega_\Lambda(K) = \min\left\{\omega_c(K) \mid c \in \Lambda^* \setminus \{0\}\right\} \quad (2)$$

where Λ^* is given by $\Lambda^* = \{x \in \mathbb{R}^d \mid x^T v \in \mathbb{Z},\ \forall v \in \Lambda\}$. When $\Lambda = \mathbb{Z}^d$, we have $\Lambda^* = \mathbb{Z}^d$ and we write $\omega(K)$ instead of $\omega_{\mathbb{Z}^d}(K)$.

Geometrically, if a set K has a lattice width of l then K can be covered by at most $\lfloor l \rfloor$+1 parallel hyperplanes given by $c^T x = \text{cst}$ with cst in the integer interval from $\min\{c^T x \mid x \in K\}$ to $\max\{c^T x \mid x \in K\}$. It is straightforward to see that $\omega(K) = \omega(\text{conv}(K))$ where $\text{conv}(K)$ denotes the convex hull of K. Here, we also suppose that K is a polyhedral convex set. K can be given either in H- or in V-representation [5]. In the H-representation, K is defined as $K = \{x \in \mathbb{R}^d \mid Ax \leq b\}$ where $A \in \mathbb{Z}^{m \times d}$ and $b \in \mathbb{Z}^m$. In the V-representation, we suppose that K is given by a list of vertices and edges. In this paper, we focus on the V-representation of K. A pre-processing must be done when K is given by an H-representation. All numbers appearing in the coordinates of vertices or in the components of vectors are supposed to have bit size bounded above by $\log s$. Moreover, the hypothesis that

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K is given is not a strong hypothesis because it is known that the convex hull of any set of integer points can be computed in linear time [10].

In the present paper, we will study the case $d = 2$ in depth and then provide a generalization to an arbitrary dimension. Moreover, we suppose that K is an integer polygon. We denote by n the number of vertices of K. The lattice Λ is also supposed to be $\mathbb{Z}^2$. In this case, the lattice width is an integer. The problem we would like to solve is the following one:

Problem (Lattice Width)
Given an integer polygon $K \in \mathbb{Z}^2$ in a V-representation, find its lattice width $\omega(K)$ as well as all vectors $c \in \mathbb{Z}^2$ such that $\omega_c(K) = \omega(K)$.

It should be noted that we are not interested in *approximate* solutions of the problem. An approximate solution of the problem might be found by a shortest vector with respect to the *dual norm* whose unit ball is the polar set of the set $\frac{1}{2}(K + (-K))$ [13, 11].

3. Planar case

We recall in this part, the result obtained in [7] via connections with the notion of digital straightness and more precisely with the notion of arithmetical digital lines [15].

The idea in [7] is based on the principle that the lattice width of K is necessarily reached for two *opposite* vertices of K. To define the notion of *opposite*, we rely on the notion of *supporting lines* well known in computational geometry [5]. A *supporting line* of K is a line D such that $D \cap K \neq \emptyset$ and K is contained entirely in one of the half-planes bounded by D. For each supporting line D, there exist at least one vertex v of K such the parallel line D_v to D passing through v is such that K entirely leads in the strip bounded by D and D_v. If s denotes a vertex of K belonging to D then s and v are called *opposite* (see Fig. 1, left). Opposite pairs are also called *antipodal* pairs. The supporting line D intersecting K along an edge is called *principal* supporting lines.

We now suppose K to be oriented counter clockwise. As in the classical Rotating Calipers algorithm of Toussaint [9], we can rotate the principal supporting lines D around the right vertex of $D \cap K$. D_v is also rotated around v to keep it parallel to D. This rotation can be pursued until D or D_v becomes another principal supporting line. Note that D and D_v are simply supporting lines during the rotation. At each position of the rotation s and v form an opposite pair of points

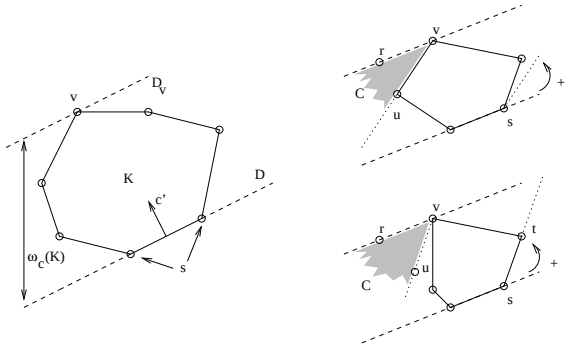


Figure 1. (left) Width $\omega_c(K)$ (c' is orthogonal to c) (right) Cone of rotations

which exactly define $\omega_c(K)$ where c is the direction of D. Hence, as depicted in Fig. 1 (right), s and v exactly define $\omega_c(K)$ for all D in a cone whose apex is v. The point r is such that the segment from v to r has exactly the same length than the opposite edge of K and the point u is either the next vertex of K after v or the point on the parallel of the line (st) such that the length of $[st]$ equals the length of $[vu]$.

After one turn around K, we have constructed at most $2n$ opposite pairs and associated cones. Hence, the number of cones is $O(n)$. Moreover, the series of cones forms a partition of all possible directions of computation for the lattice width taking into account that $\omega_c(K) = \omega_{-c}(K)$. Hence, as previously announced, the computation of the lattice width is reduced to the computation of the minimal value of $\omega_c(K)$ for each cone.

In each cone C, the computation of the minimal value of $\omega_c(K)$ is the computation of the shortest vectors for the dual norm. They are thus located at the vertices of the border of the convex hull of integer points except v inside the cone [11]. This set is also known as Klein's sail [1]. Note that to allow the possibility to find all solutions, repetitions in the convex envelope must be kept. To compute Klein's sail, we use an adapted version of the algorithm of Harvey [8] whose complexity is $O(\log s)$ arithmetic operations for constructing the sail. To bound the complexity of the search, we could also rely on the general theorem of Cook et al [4] which says that there exists $O((\log s)^{d-1})$ vertices in dimension d, a result also shown by Harvey [8] with an explicit example of the worst case in two dimension.

Thus, the complexity of the lattice width computation is $O(n \log s)$. It should be noticed that this algorithm is easily extended to the incremental case [7]. No-

tice also that with the use of the structure of Overmars and Van Leeuwen [14] for dynamic convex hull, that this algorithm might be extended to the dynamic case.

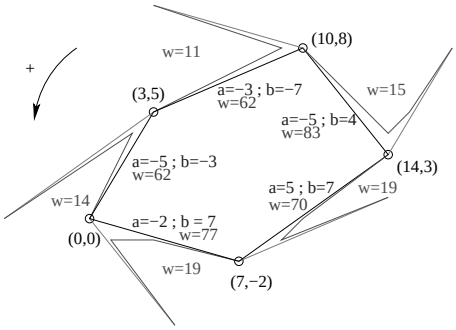


Figure 2. A complete execution of the algorithm with all sails and the thickness corresponding to the shortest vector

We now consider the example of Fig. 2. All points have been placed such that the coefficients of all the edges correspond to irreducible couple (a, b). We also provide on Fig. 2 all the sails obtained for one positive turn around the convex set. In each cone, we have depicted the sail and shown the thickness associated to the shortest vector. It should be noticed that the thickness of each edge is very high and the optimal solution could not correspond to directions given by an edge of the convex set (see the values of ω inside the convex set).

We have solved the problem when $\Lambda = \mathbb{Z}^d$. This result naturally extends to the case where K is still an integer polygon and Λ is any full-dimensional lattice of $\mathbb{Z}^d$. Indeed, there exists a linear invertible transform M of $\mathbb{Z}^d$ such that $M\Lambda = \mathbb{Z}^d$. If we consider $K_M = M^{-1}K$ then $\omega_\Lambda(K) = \omega_{\mathbb{Z}^d}(K_M)$.

4. Digital Quasi-Straight Segments

4.1 Definitions

In previous sections, we have both presented the notion of lattice width and an efficient way to compute it. We now return to the main goal of this paper that is defining a new notion of straightness. In order to provide a new approach, we first precisely describe our main goals. Contrarily to the ADSS extension [3] and the blurred segments extension [6], e want to provide a solution which can be applied both to thin digital curve and thick digital curve. The solution given in [6] also applies to the thick case but is govern only by the geometry of convex hull and only directions defined on the border of the convex hull can be used. This is unsufficient in general as shown previously in Fig. 2.

We might be attracted by the choice of a direction corresponding to the lattice width of the digital sets under study. But, this cannot simply done as is in general. Indeed, first of all, the solution of the lattice might not be unique. This is a problem and corresponds in fact to the existence of symmetries of the digital set. Moreover even if the direction of the minimal lattice width is unique, this direction is not necessarily a good solution. Indeed, the problem is that small sets are often detected as vertical or horizontal segments and thus their possible visual direction are not recovered from the minimal lattice width. To have a more robust solution, we prefer to use another criterion. To introduce it, we return in the real space for a while. In $\mathbb{R}^2$, a line is 1-dimensional. This means that the height of the orthogonal to the line is precisely 0. Thus, once we have computed all the directions corresponding to the lattice width computation, we also compute the lattice width of the digital set in the orthogonal direction. Then for each direction, the ratio between the smallest value over the largest one is the ratio of the direction. For all possible directions, we choose the one with the smallest ratio. If we fix in advance a threshold ε then the set is considered as a quasi-straight segment if and only if the smallest ratio is lower than ε.

4.2 Experiments

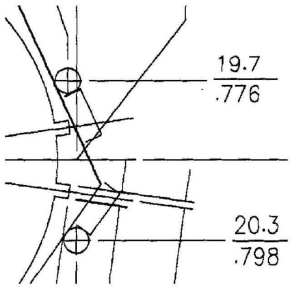


Figure 3. A raw document to be analyzed

To show the result of our method, we have done experiments on raw data sets (see Fig. 3). No preprocessing are done on the digital set, we simply consider subparts as is. As it is show in Fig. 4, the image is very irregular and noisy.



Figure 4. A zoom of the raw document

We have computed the computation of the direction with minimal ratio on the data of Fig. 4. The results reconstructed quasi-straight segments are shown in Fig. 5 where ε has been fixed to a high value ($\varepsilon = 1$) so to provide the best possible solutions extracted from the lattice width computations.

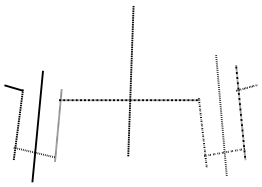


Figure 5. The reconstruction of part of the raw document (segments are longer than needed to better see their slopes)

In Fig. 5, it might be seen that the central part is not detected as a vertical one whereas it visually seems to be vertical. The reason is that on small sets, the ratio does not take into account the width value and here, it is greater than the number of points in the set. So, this can be corrected when considering only directions for which the lattice width is strictly smaller than the number of points, for instance 50% of the number of points. In that case, the central part is correctly recognize as a vertical segment.

5. Conclusion

We have presented in this paper a new approach for almost digital straight segments. Our notion is called quasi-straightness and aims at extracting the direction of a digital set either on thin or on thick digital sets. Our notion seems to be robust even if more experiments should be provided and is computationally efficient since its complexity is nearly linear in practice with no complicated data structures. Moreover, our method combines the advantages of the mathematical approach and the geometrical one because it is based on arithmetical properties of convex hulls.

Acknowledgments

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